



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte(Deemed to be University)
First Semester M.Tech. (EVT)(CBCS) Degree Examinations
January 2023

22EVT102 - VEHICLE DYNAMICS

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **One full** question from **each Unit**.*

Unit – I		Marks	BT*	CO*	PO*
1. a)	Elaborate roll of center of gravity and lumped load in vehicle dynamics.	8	L*2	1	1,2
b)	Summarize factors affecting dynamic axial loading in vehicle movement.	8	L3	1	1,2
c)	Explain newton's second law for rotational systems.	4	L1	1	1,2
2. a)	Derive expression for brake proportioning.	8	L3	1	1,2
b)	Derive dynamic load carried by front and rear wheel for low acceleration.	8	L4	1	1,2
c)	List various arbitrary force acting on vehicle.	4	L1	1	1,2
Unit – II					
3. a)	With neat diagram explain power train components with transverse weight shift.	8	L2	2	1,2
b)	Derive expression for traction limited tractive force.	8	L2	2	1,2
c)	Write a note on auto gear mechanism.	4	L1	2	1,2
4. a)	Derive expression for stop time of brake.	8	L2	2	1,2
b)	Define brake factor and derive the expression for the same.	8	L2	2	1,2
c)	With neat diagram explain working of drum type brake.	4	L2	2	1,2
Unit – III					
5. a)	Explain mechanics of boundary layer air flow around the vehicle with air friction.	8	L2	3	1,3
b)	Explain various drag components and their effect on vehicle movement.	8	L2	3	1,3
c)	Write a note on deck lid spoiler used in aerodynamic aids.	4	L1	3	1,3
6. a)	Derive slip equation and explain factor affecting slip in vehicle.	8	L3	3	1,3
b)	Explain the various types of vibrations occurring in tire.	8	L2	3	1,3
c)	Explain conduction of bias ply type of tire.	4	L2	3	1,3
Unit – IV					
7. a)	Explain the fundamentals of wheel hop resonance.	8	L2	4	1,3
b)	Explain suspension stiffness effect on riding the vehicle.	8	L2	4	1,3
c)	Elaborate the vehicle response properties.	4	L1	4	1,3
8. a)	Explain rigid body bounce dynamics observed in vehicle ride.	8	L2	4	1,3
b)	Elaborate the suspension nonlinearity effect on the ride.	8	L4	4	1,3
c)	Write a note on roll of tire and wheel assembly in vehicle ride.	4	L1	4	1,3
Unit – V					
9. a)	Elaborate the anti-dive suspension geometry used in vehicle.	10	L3	5	1,2
b)	Explain the roll center analysis of suspension used in vehicles.	10	L2	5	1,2
10. a)	Elaborate independent suspension roll center of vehicle.	10	L2	5	1,2
b)	Explain front wheel steering system forces and moments.	10	L2	5	1,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
